

Regression Analysis on Weekly Vehicle Thefts and Limitations in Capturing Seasonal Effects*

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Every year, a significant number of vehicle theft reports are filed by the Police Department in California, USA. This project aimed to apply simple linear regression analysis to explore week-by-week trends of vehicle thefts that were reported in 2024 in San Francisco. The analysis results suggested that there is an overall decrease in thefts in 2024 compared to past years (2020-2023). However, the regression model was not fit to address any seasonal patterns in the data. The model failed to capture the fluctuations in the data and provided no significant linear trend between the two variables. This highlights the gap in knowledge of choosing the predictor (week) and limitation of the linear regression model in detecting seasonal patterns. The fluctuations may be due to various factors involved, such as holidays or increased monitoring affecting the decline in the number of vehicle theft reports. Future analysis using non-linear models is required to capture the seasonal trend in the data.

1 Introduction

In the United States, the issue of vehicle theft has significantly impacted residents for many years. Incidents of stolen vehicles, break-ins, and parts theft have been increasing rapidly. However, in 2024, San Francisco, California, saw its first significant decrease in the number of such cases (Fang 2025).

Figure 1 displays the sudden decline in the number of motor vehicle theft incidents in San Francisco for the year 2024, compared to previous years starting from 2020. Many factors related to the resource management and allocation by the San Francisco Police Department (SFPD) could have resulted in the decline in the number of vehicle thefts in 2024. While researchers often emphasize the overall crime trend and spatial data for coming up with theft

*Project repository available at: <https://github.com/shyamaku/MATH261A-project>.

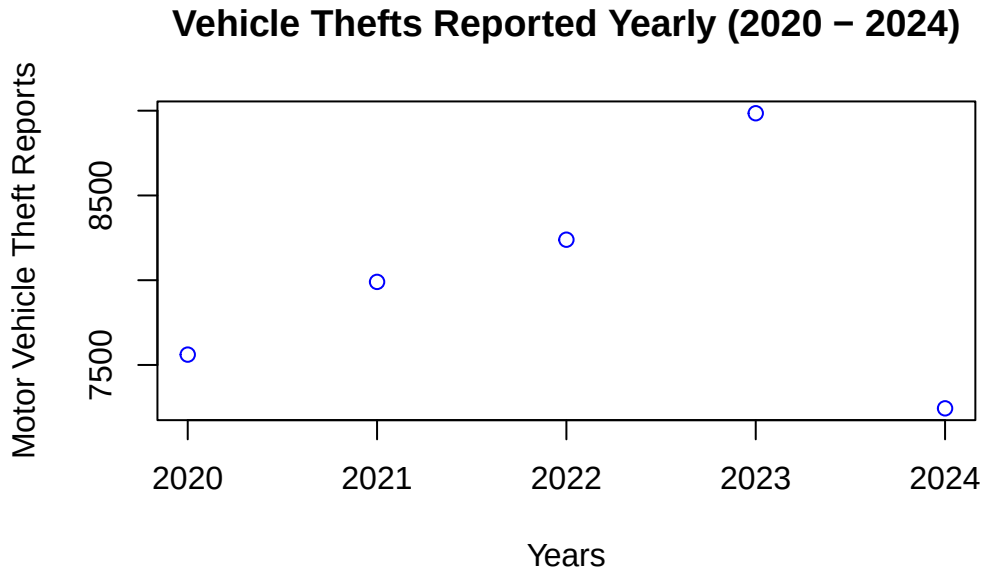


Figure 1

prevention measures, there has been less emphasis on the seasonal trend of vehicle thefts. The motivation of this project was to explore whether a simple linear regression model captures the decline in theft reports and fluctuations, such as spikes, in the summer weeks and a decrease in the winter weeks. Through this study, the expectation was to provide significant results with the aim of assisting the SFPD with better resource allocations during those periods to further reduce the crime rates.

The project analyzed trends using scatter plots and applied simple linear regression to interpret the relationship between weeks of 2024 and vehicle theft reports. It also examined the fit of the model using hypothesis testing (t-test and ANOVA) and the residual vs fitted plot. The model detected a linear relationship between weeks and the number of vehicle thefts, which cannot be said with certainty, as the number of weeks passed cannot predict the number of vehicle thefts in a week with accuracy. Additionally, the model failed to identify any fluctuations in the data, which has been discussed as a limitation of a linear regression model.

The remainder of this paper is structured as follows: Section 2 introduces the data used in this analysis. Section 3 describes the simple linear regression model used for the purpose of answering the research question. It also explains the significance of using ANOVA and residuals in assessing the fit of the model. Section 4 explores in depth the results of the analysis and examines if the assumptions of the model being a good fit hold true. Section 5 discusses the interpretations based on the research question and whether the model is efficient in analyzing the seasonal patterns, if any. Additionally, it discusses the limitations of the model for the

given data and scope of future research.

2 Data

The dataset used in the project was obtained from DataSF, which is an open data portal (DataSF 2025). It is considered a reliable source of data as the incidents are recorded through official police reporting systems. However, it may contain some wrong classifications due to human errors at the time of reporting, and there might be cases where the thefts have not been reported. Additionally, it does not contain any confidential information related to the incidents. The dataset comprises 969326 records of various incidents (e.g., Fraud, Assault, Motor Vehicle Theft) captured from 2018 to the present. Each row describes the date, time, year of an incident, along with incident description, ID, category, police district, analysis neighborhood, resolution, and other incident area-related fields.

For the scope of this analysis, the data was filtered to include only records of “Motor Vehicle Theft” for the year 2024. Only one year of data was chosen with the intent that it would reflect the most recent condition of the thefts in the city of San Francisco. The data was processed in a weekly format, as the number of theft reports daily might not significantly express the fluctuations, while a monthly analysis might miss any granular-level trend in the data. As per the initial assumptions, the data was chosen to detect patterns that might assist in further theft reduction by examining one year of data and identifying week-by-week variations. The key variables used in the regression analysis are mentioned below:

Key Variables	Description
week (numerical)	Week of the year. Extracted from the Incident Date field.
num_theft (numerical)	Aggregated number of vehicle theft reports for each week of the year.

While for the current analysis only one year of data was chosen, confirming seasonal patterns would require examination of multiple years of data.

3 Methods

This project adopted the simple linear regression model with week of the year as the predictor X_i and number of vehicle thefts as the response Y_i , where $i = 1, 2, \dots, n$ and n is the number of records. The model can be represented with the following equation:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

In this model, β_1 represents the slope, which describes the change in the number of vehicle thefts as the weeks increase. β_0 represents the intercept, which describes the number of vehicle thefts at week 0, and ϵ represents the random error, which describes the unexplained variation of data. The error term ϵ should satisfy the assumptions of a linear regression model, which are: The error terms were assumed to satisfy linearity, be independent, be normally distributed, and have equal variance.

Next, after fitting the model as per the above equation, the project analyzed the model's fit using the residual values. Residuals are the difference between observed and predicted values, written as,

$$e_i = Y_i - \hat{Y}$$

The assumptions of the errors are discussed in the following section.

The analysis was implemented using the R programming language (R Core Team 2024) and packages including (Wickham et al. 2023),(Wickham 2016)

4 Results

This section presents the result of the analysis conducted to examine whether motor vehicle theft reports with week as a predictor displayed any week-by-week linear trend or provided any insight into seasonal fluctuations. A scatterplot was used to visualize the relationship between weeks and the number of vehicle theft reports. Figure 2 showed a negative relation between the two key variables. Data fluctuations were observed in the scatterplot. Weeks 23-37 showed high numbers of vehicle theft reports.

After fitting the model using the number of vehicle thefts and week, the model provided no information about the fluctuations noticed in the scatterplot. The estimated slope parameter was $\beta_1 = -0.953$. In other words, for each additional week, there was an average decrease in the number of vehicle theft reports by ≈ 0.95 . The estimated intercept was $\beta_0 = 164.572$, which explained the expected number of thefts at the start of the year. The R-squared value: 0.3279 explained $\approx 33\%$ of the variation in the theft report numbers per week. The predictor did not cover a significant amount of variation in the data.

To assess the fit of the linear regression model using the t-test, hypotheses were defined. The null hypothesis $\beta_1 = 0$ (no linear relationship between the number of vehicle theft reports and week of the year) was compared with the two-sided alternative hypothesis $\beta_1 \neq 0$. The t value = -4.9389423 and p-value = 9.1749221×10^{-6} from the summary of the linear regression model showed that the week of the year was a statistically significant predictor for the estimated decrease in the number of vehicle theft reports. However, this could not be interpreted as such because it was already established that in 2024, the number of vehicle thefts had decreased significantly, and the decrease could not directly depend on the increase in weeks. As time progresses naturally, the model could show statistical significance, which made the week number a bad predictor, and no relationship could be mentioned with certainty between the

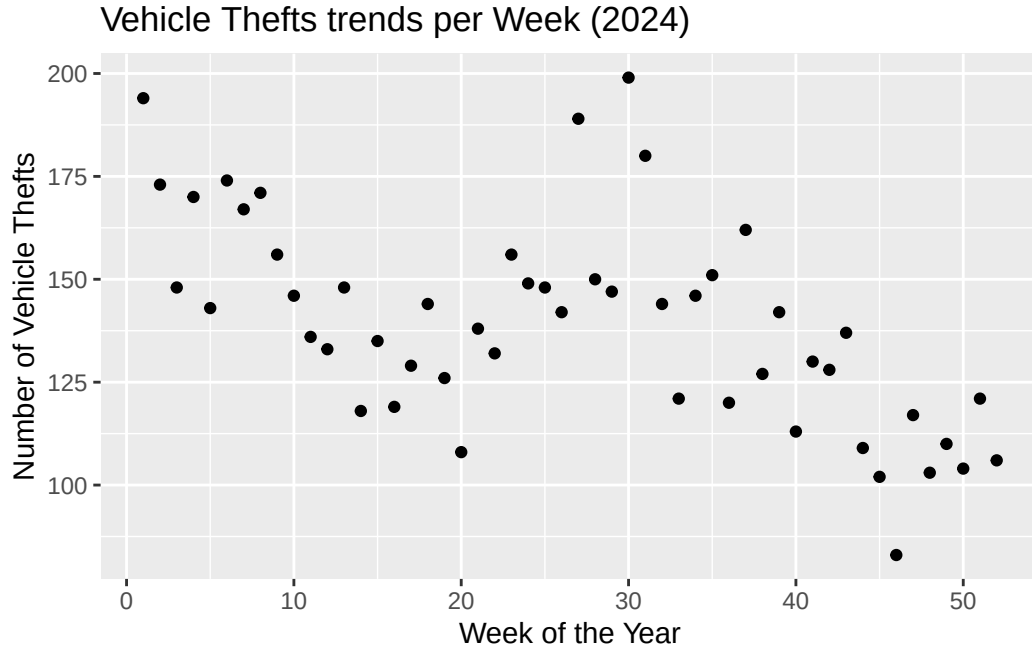


Figure 2: Scatter plot of week in x-axis and number of vehicle theft reports in y-axis.

two variables. In this case, further evaluation is needed with multiple regression models, and thus, the null hypothesis could be accepted.

Another way to assess the fit of the linear regression model was to examine the residual vs fitted value plot and assess if the assumptions of a linear regression model hold for the analysis. Figure 3 examined the residuals to assess model fit. From the scatter plot, the linear regression model assumptions were assessed. Except for error terms having equal variance, no other assumptions were satisfied. Linearity - The relationship between the two variables should be linear. However, the plot suggested that there might be some amount of linear relationship between the number of vehicle theft reports and weeks, as there are no visible arcs or curves in the plot, but due to some points forming waves, it is difficult to interpret the same with certainty. Equal Variance - The error terms had equal variance, as there was no major spread visible in the data points of the residual plot. Independence of errors - The error terms were not independent, as an increase and decrease in the residual points could be observed forming waves with the increase in the fitted values. Normality of Errors - The error terms might not be normally distributed, but to verify that, a quantile-quantile plot would be needed. From the above interpretations, it could be said that the model is not a good fit for the chosen dataset and was unable to satisfy the research assumptions made initially.

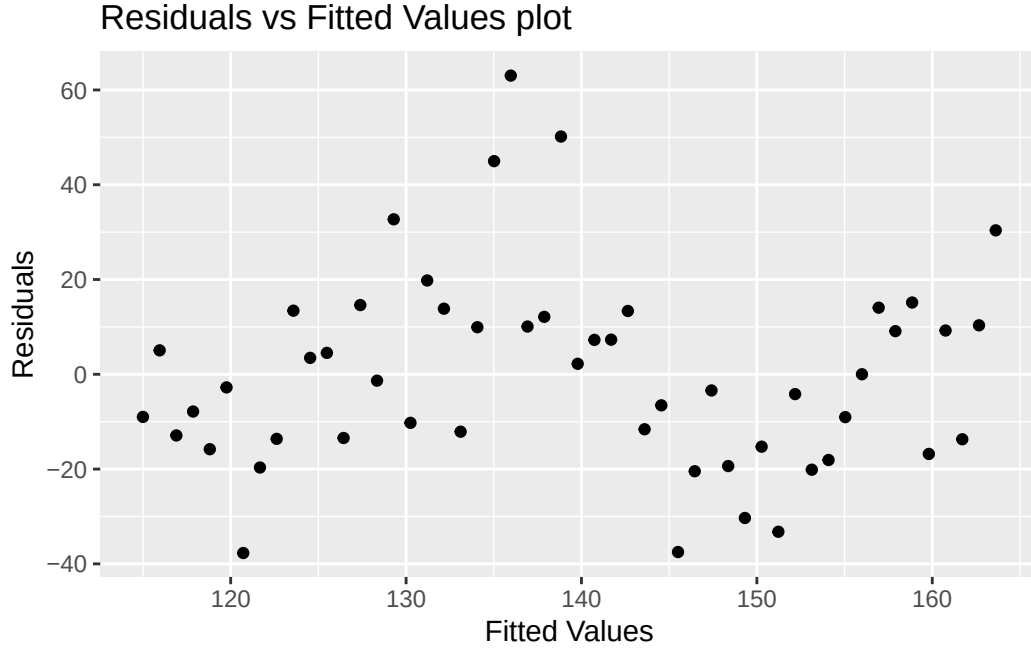


Figure 3: Scatter plot of fitted values (x-axis) and residuals (y-axis)

5 Discussion

The analysis of vehicle thefts and the week of the year 2024 revealed no linear patterns, as the predictor chosen was an inadequate parameter in identifying any linear patterns from the data. No interpretation can be made with certainty with the chosen predictor and response variable, as after assessing the residual vs fitted model plot, it can be said that the linear regression model does not fit the data. As the errors are not independent, there might be non-linear patterns that the linear regression model is not fit to expand. Even though the hypothesis testing using the t-test showed that the predictor is statistically significant, week is a time parameter, and thefts would not be influenced by the progression of weeks.

According to the initial assumption, Figure 2 indicated that there were fluctuations in the data as the weeks progressed. However, the analysis presented the key limitation, using a simple linear regression model was not adequate to answer the research question, as showing any non-linear seasonal patterns is not in the scope of the model. There can be various factors influencing the increasing and decreasing patterns of the number of vehicle thefts, such as an increase in police patrols in the middle to latter half of the year or implementation of automated drones and cameras to increase surveillance. As mentioned previously, this highlights a knowledge gap that should be addressed in future research.

For further inspection of the assumptions made initially, the analysis should be done with

multiple regression models or time-series models using multiple years of data to assess if the seasonal trend is valid for each year. Analysis with the addition of a few other key predictors, like police districts or incident time (day or night), should be used to analyze the seasonal trends better with multiple regression models. Time series models should be explored for a better analysis of the data to explore the non-linear patterns. This could be helpful in exploring different trends of the number of vehicle thefts across weeks, which would prove significant towards the real goal of assisting the SFPD with enhanced planning and reducing the theft rates further.

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